

CIS-4327 Database Admin

Spring/2024 – 10 January 2024 – 29 April 2024

Instructor: William Nash

Phone: (972) 372-4557

Email: nash@txwes.edu

Course modality: Online/Face-to-Face

Meeting Times: M, W, 6:00 PM – 7:15 PM Armstrong Mabee Business Cntr, Room 214

Office: McFadden 103

Office Hours: Times by appointment

Course Introduction

Welcome to Database Fundamentals! I'm excited to help you build a strong foundation in database concepts using Microsoft Azure cloud services. Understanding how to leverage the robust data storage, processing, security, and analytics capabilities of Azure will give you an advantage in our increasingly data-driven world.

Through this course, you will gain relevant skills for managing modern data workloads. We will explore relational vs. non-relational systems, cloud database deployment, data warehousing, big data processing, and much more. Importantly, you will get hands-on experience with Azure SQL, Cosmos DB, Blob storage, data pipelines, and other key services.

A significant focus will be preparing you to pass the Microsoft Azure Data Fundamentals certification exam (DP-900) which validates your abilities in provisioning and operating core Azure data services. Previous students have used this beginner-level certification to stand out and get jobs in cloud database administration, data analysis, system administration, and more.

I'm committed to making sure you finish this course with a balanced mix of conceptual knowledge and practical ability. My teaching style is to explain database/Azure concepts clearly through lectures and demos, while also letting you get direct hands-on practice. Learning by doing cements your abilities to apply these data skills in real job roles.

Let the curiosity and questions flow as we tackle more advanced data topics each week! By the end, you will gain valued Azure data skills and have the certification to prove it. I'm excited for the data journey ahead this semester that will open up new career opportunities. Let's do this!

Course Description

This course provides an introduction to database concepts with a focus on cloud-based database services in Microsoft Azure. Students will learn the foundations of relational and non-relational data, Azure data workloads, security, privacy, and monitoring. Hands-on labs give practical experience with Azure data storage options including SQL databases, Cosmos DB, storage tables/queues, and file storage.

By the end of the course, students will gain the skills needed to provision Azure data resources, ingest and query data, and implement data security. Along with learning key capabilities of Azure data

services, students will prepare for the Microsoft Certified Azure Data Fundamentals certification exam (DP-900).

Topics mapped to the DP-900 exam include:

- Azure data fundamentals
- Relational vs non-relational data
- Monitoring data workloads
- Privacy and Compliance
- Data ingestion
- Data warehouse solutions
- Data security
- Azure data solutions

This course combines conceptual readings with technical hands-on labs working in Azure to provide job-relevant cloud database skills as well as DP-900 certification. The certification validates abilities in implementing core Azure data services which is invaluable for data-driven roles.

Course Learning Objectives

Program Goals

Describe options and tradeoffs between relational and non-relational Azure data services to meet solution requirements	Explain when to appropriately apply relational vs. non-relational data systems to meet business objectives
Provision and deploy scalable Azure SQL and non-SQL databases	Display competency in deploying fully managed databases in Azure cloud
Import, export, transform, and query data using tools like Azure SQL DB, Synapse Analytics, Cosmos DB, and data pipelines	Display competency in deploying fully managed databases in Azure cloud
Implement speed, performance, replication, and disaster recovery in Azure data solutions	Shows aptitude for ingesting, transforming, and analyzing data in Azure environments
Ingest real-time and batch data streams with tools like Event & IoT Hubs, Logic Apps, and Azure Functions	Implement scalable, secure, performant, and cost-optimized data solutions
Monitor and optimize costs, security, and performance of Azure data resources	Monitor Azure data workloads using built-in tools and logs
Use built-in reporting to analyze utilization of Azure data services	Pass the Microsoft Azure Data Fundamentals certification (DP-900) exam

Required Materials

Textbook(s)

Author: Gaurav Aroraa
Title: Microsoft Azure Data Fundamentals Certification Guide
Publisher: Packt Publishing

Year: 2021
ISBN: 9781801814065

Author: Daniel Crawley
Title: Exam Ref DP-900 Microsoft Azure Data Fundamentals
Publisher: Microsoft Press
Year: 2021
ISBN: 9780137488339

Software/Apps/Equipment

VirtualBox: Software is needed to run different types of databases.

Laptop or Desktop Computer

Linux Centos: This will be installed as a Virtual Machine in VirtualBox.

MariaDB/MySQL: Added to Linux

MongoDB: Added to Linux

Microsoft SQL 2019: Added to Windows VirtualBox

Microsoft SSMS: Added to Windows VirtualBox

Instructional/Learning Methods and Technologies

The learning methods used in this course include the following:

- ☒ Discussion forums and/or wikis
- ☒ Synchronous (real-time) virtual meetings with peers and/or the instructor
- ☒ Streaming video
- ☒ Quizzes
- ☒ Exams
- ☒ Collaborative assignments
- ☒ Independent homework assignments
- ☐ Other:

Student Workload Expectation

This course is a 3-credit hour course, which means that in addition to the scheduled lectures/discussions, students are expected to do at least 300 minutes of course-related work each week during the semester. This independent learning includes things like completing assigned readings, participating in lab sessions, studying for tests and examinations, preparing written assignments, completing an internship or clinical placement requirements, and other tasks that must be completed to earn credit in the course.

Course Requirements

Submitting and Returning Assignments

Assignment Submission

- Submit all your assignments via Canvas.
- Use the standard Microsoft® Office programs such as Word, PowerPoint, and Excel (available to you through your txwes.edu Office 365 account) to complete your assignments unless assignment directions specify that you should use a different file format.

Assignment Feedback

- I will provide feedback on assessments within 2 days after submission.
- My feedback will be provided as a written response via Canvas.
- Your grades will be posted to the Canvas Grade Center.

Communication with the Instructor

- To contact me with questions and concerns about this course, post to the Q&A discussion forum, which I will check throughout the day. I will address questions posted after 9 p.m. on the following morning. You should also check the forum regularly, as others in the class may be able to answer your question before I do.
- Send questions of a personal nature to me via email. I will respond to emails within 24 hours.

Quizzes and Exams

- Quizzes will be administered online while using the Respondus lockdown browser.
- The final exam will be a proctored exam at ComputerMinds, INC testing center.

Evaluation & Grading

All grades will be made available to students in Blackboard.

Your final grade in this course will be based on the percentage of points that you receive out of the total possible points for the course. Grades will be determined according to the following scale:

90%-100%	A
80%-90%	B
70%-79%	C
60%-69%	D
59%-0%	F

The weight/point distribution is as follows:

Category	Weight
Homework	10
Quizzes	30

Final Exam	60
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Course Policies

Academic Integrity

- Cheating, plagiarism (submitting another person's material as one's own), or completing assignments for another person who will receive academic credit are not permitted. This includes the use of unauthorized books, notebooks, or other sources in order to secure or give help during an examination, the unauthorized copying of examinations, assignments, reports, or term papers, or the presentation of unacknowledged material as if it were the student's own work. Disciplinary action may be taken beyond the academic discipline administered by the course instructor.
- [Read the University's Academic Integrity Policy.](#)

Instruction/Attendance definitions

- **Asynchronous** – students learning asynchronously are not attending class in-person or virtually. Asynchronous students will be provided recordings of class lectures (audio).
- **Face-to-Face** – faculty deliver live instruction in the classroom. Subject to the restriction noted above (#4) students may choose to attend a FTF class in-person or virtually.
- **In-person**- students are physically present in the classroom during the designated face-to-face (FTF) class lecture.
- **Virtual or virtually** – students attend class using an online platform such as Collaborate or Teams. Students are expected to be online and participate in class at the designated meeting time for the class period.

Face-to-face Class Attendance

Regular and punctual attendance at all scheduled classes is expected of all students. Students are required to notify instructors before any missed class and will be held responsible for all class and laboratory assignments.

The maximum number of authorized absences during one semester is:

- Four for a class that meets once per week
- Five for a Monday/Wednesday or a Tuesday/Thursday class
- Seven for a Monday/Wednesday/Friday class

Additional authorized absences may be granted on an individual basis when extreme circumstances warrant.

A student who has been absent from any regular examination and has satisfied the instructor that the absence was due to serious illness, or some other unavoidable cause may take a special examination at the discretion of the instructor concerned.

Students who have the following number of non-consecutive unauthorized absences may be dropped from the course by the instructor:

- One unauthorized absence for classes that meet one time a week
- Two unauthorized absences for classes that meet two times a week

- Three unauthorized absences for classes that meet three times a week

Online and Hybrid Class Attendance and/or Participation

Students enrolled in distance education courses must demonstrate academic engagement throughout the course to be counted as present.

Academic engagement means actively engaging with course content. Examples of academic engagement include attending a synchronous session, interacting with your instructor, participating in an online discussion, completing interactive tutorials or web-based assignments, and completing assignments or exams.

Academic engagement does not include the act of logging into Canvas, nor does it include participating in academic counseling or advising.

Students must demonstrate academic engagement a minimum of once each week to be marked present for that week. During the first week of class, all students will be required to complete either a grade-based or submission-based verification of academic engagement.

Faculty Initiated Student Withdrawal from a Course. When a student accumulates consecutive unauthorized absences equal to the number of days the class meets per week (1, 2, or 3) AND has not communicated with the instructor about the absences, the instructor must drop the student from the class using the Faculty Initiated Student Withdrawal process within 10 calendar days from the last date of attendance or give a grade of incomplete (I) as appropriate.

The Registrar will record a grade of "DP" for the course on the student's transcript. The Registrar will record a grade of "WF" (failure) on the student's transcript if the instructor drops the student after the date for receiving a grade of "DP" as specified in the University catalog. Note: Per the Office of Financial Aid, students who withdraw from a course cannot be re-enrolled in that course during the same semester.

[The complete Attendance Policy is available in the Catalog.](#)

Other University Policies

Selected University policies are available in the [Canvas TXWES_STUDENT_HUB](#). The current Texas Wesleyan University [Catalog](#) and [Student Handbook](#) contain all University policies.

Announcements:

Announcements will be posted in the course as needed to keep students informed of changes in schedule or points of clarification for the course. Students should check the announcements each time they enter the course.

Schedule

Week Topic

Week 1 (Jan 10, Jan 15) Introduction to Azure Data Fundamentals

Week 2 (Jan 22, Jan 24) Relational vs. Non-Relational Data

Week 3 (Jan 29, Jan 31) Provisioning Relational Data in Azure Hands-on Lab: Deploy Azure SQL Database

Week 4 (Feb 5, Feb 7)

Azure Non-Relational Data Services Hands-on Lab: Cosmos DB

Week 5 (Feb 12, Feb 14) Data Ingestion and Processing Hands-on Lab: Data Factory Pipelines

Week 6 (Feb 19, Feb 21) Real-time Data Streaming in Azure
Hands-on Lab: Event Hubs

Week 7 (Feb 26, Feb 28) Data Warehousing Solutions Hands-on Lab: Synapse Analytics

Week 8 (Mar 4, Mar 6) Monitoring Data Workloads Hands-on Lab: Azure Monitor

Week 9 (Mar 18, Mar 20)
Securing and Protecting Data
Hands-on Lab: Encryption

Week 10 (Mar 25, Mar 27) Compliance and Privacy Hands-on Lab: Compliance Tools

Week 11 (Apr 1, Apr 3) Review Course Learning
DP-900 Practice Exam

Week 12 (Apr 8, Apr 10) Azure Data Services Overview DP-900 Practice Exam Review

Week 13 (Apr 15, Apr 17)
Final Exam Preparation

Week 14 (Apr 22, Apr 24) DP-900 Certification Exam